Blossom Buddy

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Abstract: This paper presents a novel interaction system combining nature and technology, and self-development through a motivational interactive system. A user engages with the real plant companion which responds with the user's daily self-development tasks. Moreover, for enhancing motivation the system offers rewards after completing a task. As some sensors continuously monitor the plant's health, it would show the water, sunlight, and nutrition levels on the screen. A user can see his or her progress through the interactive digital screen of Blossom-Buddy. It is not only a tracker it works like a mirror companion. This abstract represents a user-centered design, iterative development, and a motivational reward approach which makes Blossom-Buddy an innovative and engaging system for personal self-development.

1.Introduction:

1.1 A brief overview of the concept:

It is not always easy to stay motivated and do everyday tasks or give up on bad habits. The concept of Blossom-buddy is a rare combination of nature and technology that aims to make personal growth more innovative and attractive. Imagine having a real plant buddy who will respond to your daily self-improvement activities. Every single day, you can select your specific set of tasks and whenever you finish a single task you will get a reward for your Blossom-buddy. After completing the task for the whole set, you will get a special reward. Furthermore, you will get a surprise reward after each three or four days. This reward will motivate you to complete your next task or the set of tasks for that specific day. This display will make your Blossom Buddy livelier, and the virtual presentation of the plant helps you to engage more. Some sensors would have used for the real plant to continuously count the water level, nutrition level, and the amount of sunlight needed for a healthy plant.

1.2 The reasons behind choosing this concept:

Our team believed Blossom-buddy is not only a tracker, but it was also a friend of a user who would show regular effort to be a better version of oneself. Significantly, it could give a rapid boost to your self-growing journey by making an emotional engagement, showing the daily progress history, and the innovative motivational reward approach.

2. Prototype Development Work:

An early design representation of a final artifact is known as a Prototype (Buchenau and Suri, 2000). Blossom-buddy's prototype followed a strategy where emphasizing collaboration, iterative development, creativity, and brainstorming are perfectly involved. The prototyping structure had different key phrases like formulating the concept, identifying appropriate prototyping concepts and techniques, producing early prototypes, and producing the final version.

2.1 Formulating the Concept:

We arranged several brainstorming sessions before finalizing the idea of the blossom buddy. The shortlisted ideas were VR cooking tool, clothes app, training app, smart window, crank-powered productivity device, smart food container, blossom buddy (self-improvement motivator)

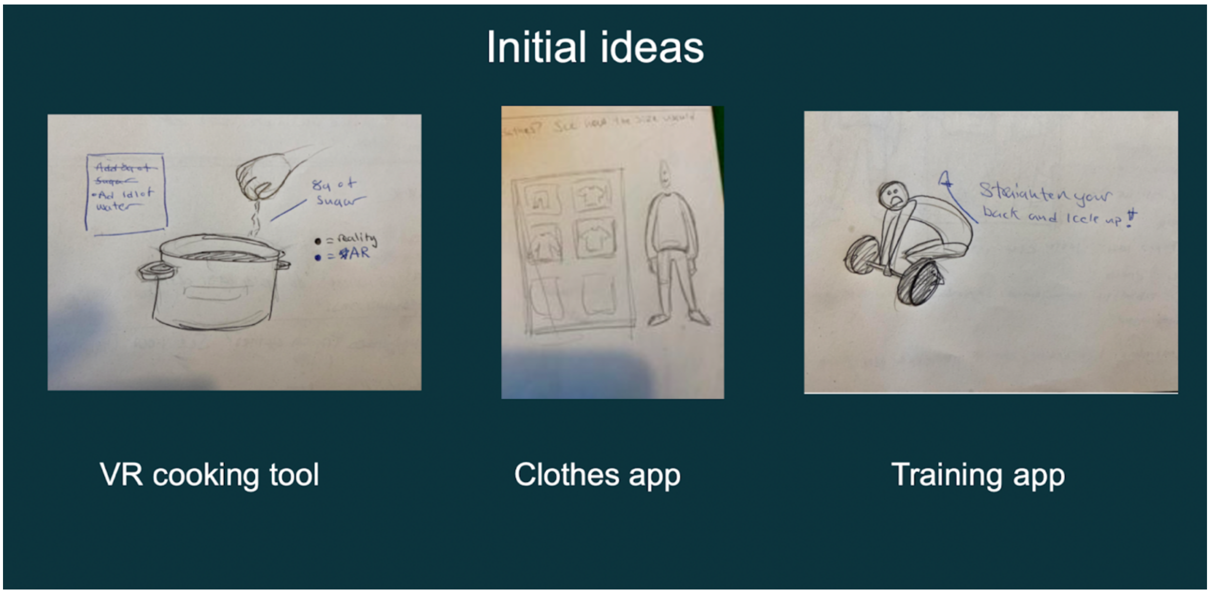

Figure 1: Ideas 1

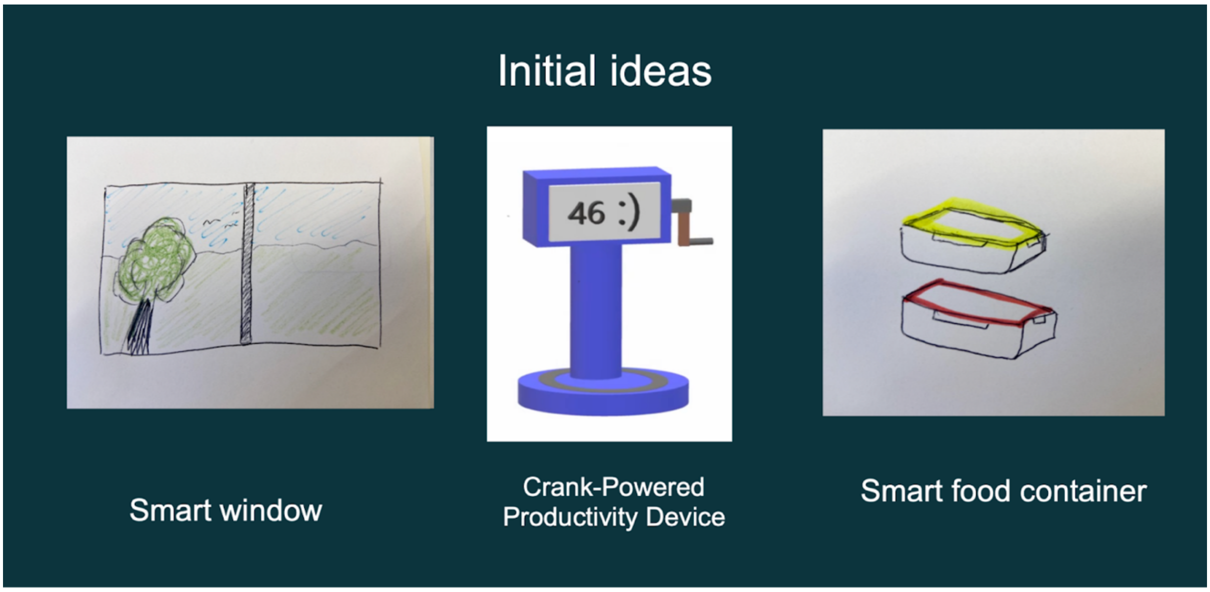

Figure 2:Ideas 2

Figure 3:Blossom-Buddy

The blossom-buddy was completely user-centered interactive system where the user's demand and the user's interactive process had been prioritized the most. First and foremost, we wanted to make a tangible interactive system which is easy to interact with in different age groups of people. We chose the theme of personal growth and improvement. It helped us to collect information and different ideas on our main theme. Moreover, we found out what the users' expectations from such an interactive system and about their pain points. Additionally, we also analyzed the plant, how it could embed into this interactive system and how it could be able to make a user emotionally engage with the system. We tried to have a clear understanding of the users with different age groups and their preferences as whatever we did, we did it for users' satisfactions. With our collaborative decisions, we chose Blossom Buddy as our main concept. Among all the ideas we found the concept of the Blossom buddy seems more unique and emotionally engageable to the user.

2.2 Identifying Prototyping Concepts and Techniques:

We selected the Experience prototype as our main prototyping method. As Experience prototyping was something that a designer, client, or user could experience very closely, they did not make decisions based on other experiences. It also included some techniques such as storyboarding, scenarios, sketches, and videos. (Buchenau and Suri, 2000). With the High-fidelity prototype, we tried to make the system more intractable by using the Experience prototyping method. Houde & Hill (1997) said that designers represented one of the three dimensions; look and feel, role, or implementation through their prototypes. In their model, integrated prototypes can also be utilized to explore a balance of aspects between all three dimensions. In blossom-buddy we created a prototype to explore the balance between the three dimensions look and feel, role or implementation (Blomkvist and Holmlid, 2007).
The primary sketch and low-fidelity prototype of Blossom Buddy represented a perfect example of look and feel. It helped designers to represent their ideas in reality. Look and feel included user interface design, and user and sensory engagement. The role was the functionality of the blossom buddy of a user's life. For example: how the users interacted with the system and the role of Blossom Buddy in their life. Implementation focused on the technical part of Blossom Buddy. For instance, the integration technology such as Figma. We used Figma a popular design software for our high-fidelity prototype. Through Figma, we tried to make our blossom buddy more visually appealing and relatable to the functionality.
For user testing, we used the Wizard of Oz (abbr. Woz) prototyping technique. Wizard of Oz (abbr. WoZ) prototyping is a technique used to test a concept/prototype that may not exist yet but appears to be functional because someone is controlling it behind the scenes." (Lee, 2008)

2.3 Producing Early Prototypes (Low-Fidelity):
As low-fidelity prototypes were very cheap and easy to use at the same time it was easy to iterate on low-fidelity prototypes therefore we had chosen a paper-based low-fidelity prototype sketching prototype for primary validation of our concept. Johan & Stefan (2007) said that prototype, the method that is develop for visual representation in design can be used. Also sketching is such a method that resembles a prototype in many ways.

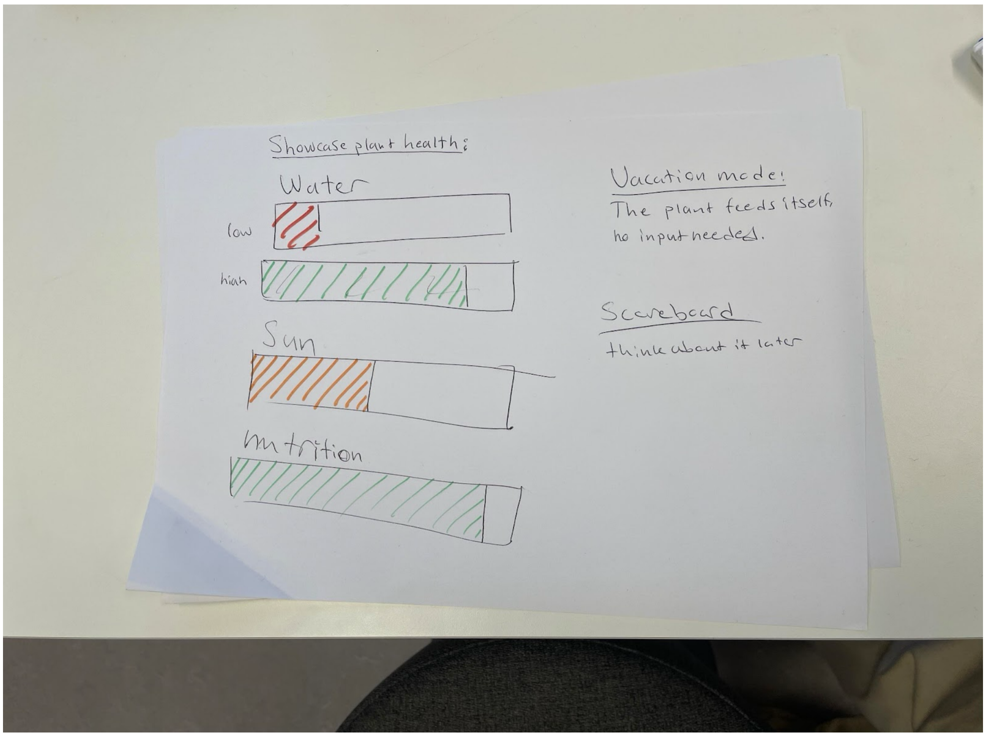

Figure 4:initial progress bar of blossom buddy

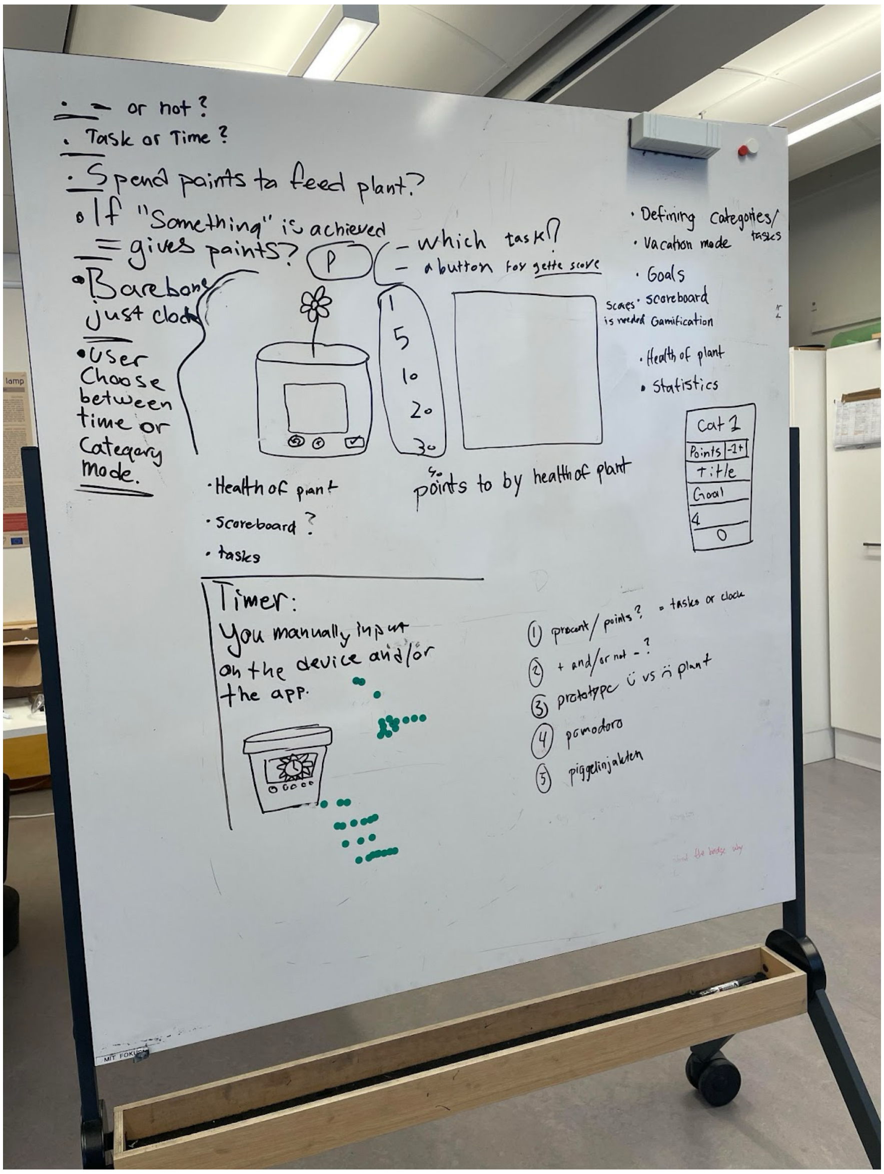

Figure 5:Digital interface prototyping of blossom buddy 1(low fidelity)

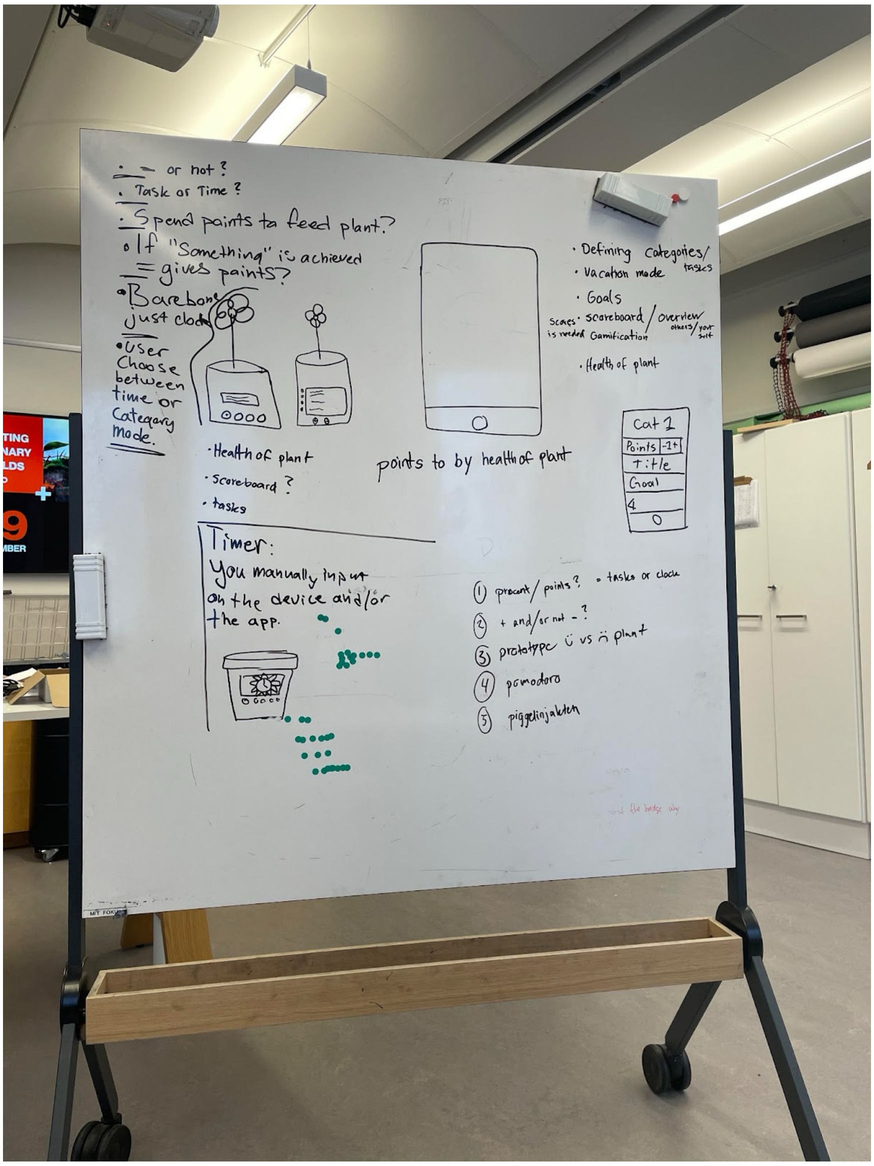

Figure 6:Digital interface prototyping of blossom buddy 2 (low fidelity)

This was the first step of the initial reflection of our idea of blossom buddy. After sketching the idea and further brainstorming we had the chance to redefine and iterate our design concepts.

2.4 Producing the Final Version (High-Fidelity):
In the early producing prototype, we thought we would prototype a display for the plant port and an application for interacting with the screen for remote interaction.

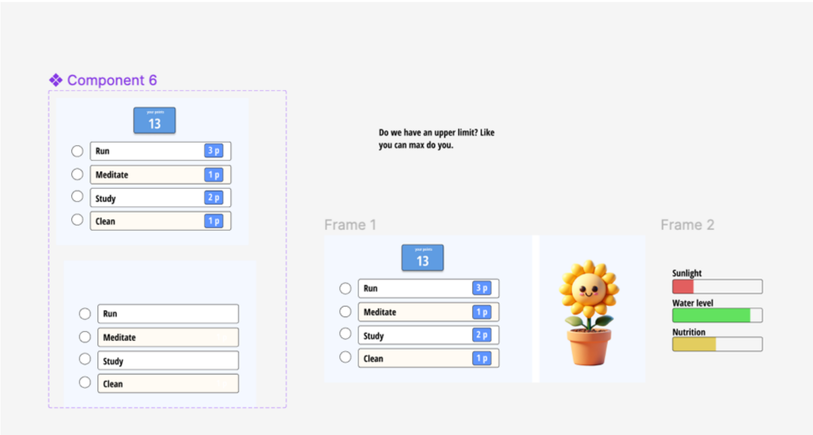

Figure 7:Primary digital interface design 1 (High-fidelity)

In the plant pot, we had embedded some sensors that would monitor its light, nutrition, and water level continuously. When a user would complete his task, the system would feed the plant automatically if needed. We had only a progress bar where users can see the health condition of the plant, for example, water level, nutrition level, and sunlight condition. We got some feedback from our professors to add some new features to make the interface attractive and game-like. We used this feedback by giving the virtual flower a personality. For example:if the plant got proper light, nutrition, and water it would happy as well as it would display its happy face. In contradiction, if the user skipped his tasks for several days and the plant does not get proper water, nutrition, and light it would show a sad face.

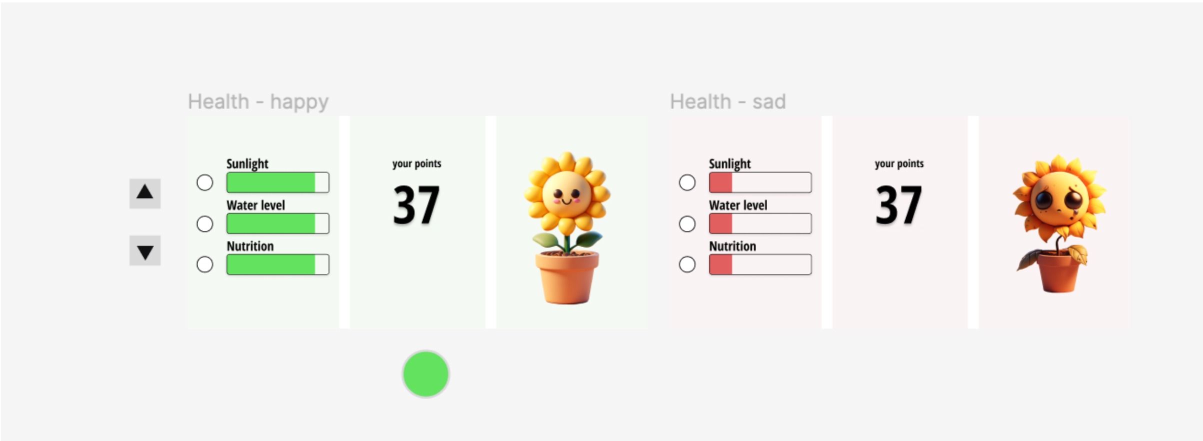

Figure 8:Primary digital interface design 2 (High-fidelity)

Moreover, a score-counting system was used in our early producing prototype. If a user completed his specific task the specific amount of score would be added to their score count and after earning a specific amount of points the system would automatically feed the plant. However, the automatic feeding of the plant process had proven to hinder our system. What if the plant had enough water, nutrition, or light, it did not need anything for the day and only because the user had accomplished his goal it would feed unnecessary water, light, or nutrition? This excessive care can cause great harm to the plant instead of taking care of the plant's growth. Furthermore, we used a physical bottom in our display to make it more convenient for the user to use.

After several supervisions, we added some new features to our interactive system.We tried to refine our system and the biggest challenge was to make it more interesting for the user and keep them bound the users with the system for a long time to avoid boredom. We rethought and redesigned our prototype to make users bound to it. After the iterative process of brainstorming, we added a motivational reward system to attract all age groups of users.

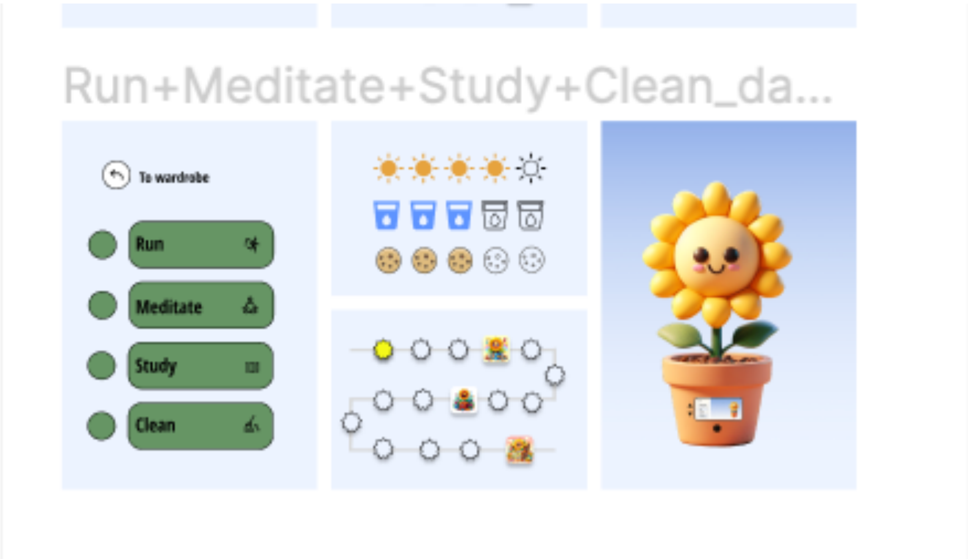

Figure 9:Final digital interface design 1 (High-fidelity)

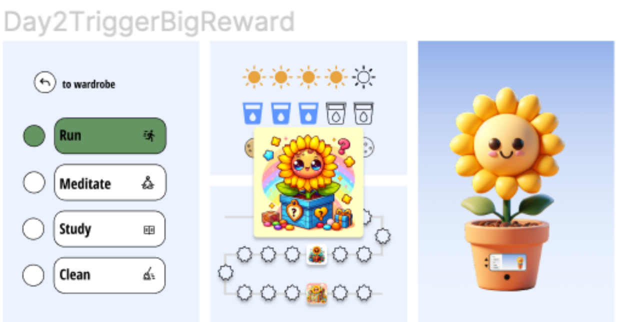

Figure 10:Final digital interface design 2 (High-fidelity)

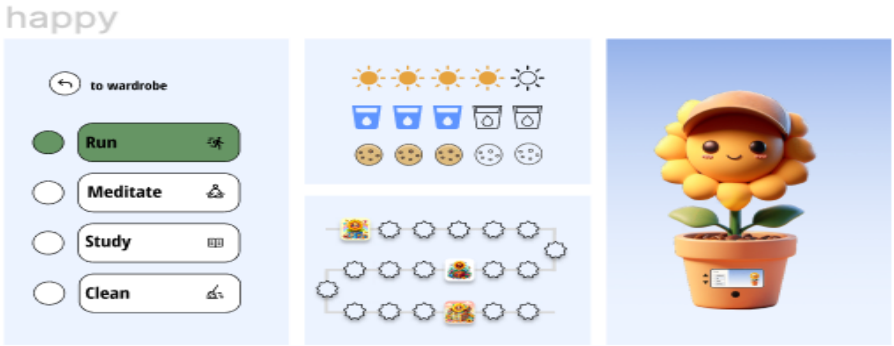

Figure 11:Final digital interface design with special reward 3 (High-fidelity)

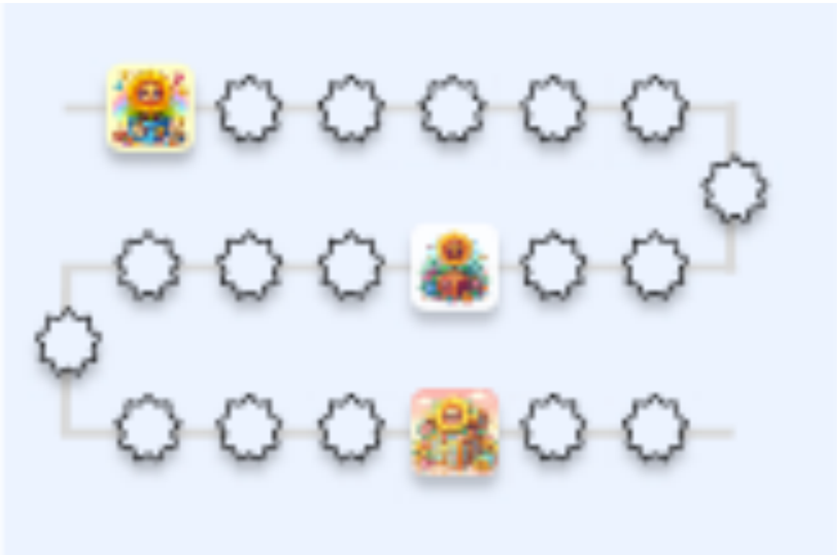

Figure 12:Final progress bar of digital interactive design

For instance, after completing a full set of tasks a user would get a reward for its virtual plant. It could be hat, new background, or clothes for his plant. If the user would notice the reward bar, they would see some star which would keep their everyday task history. If the user completed one task the star would fill a bit, slowly with every task it would increase its level. Finally, it would be full of yellow color. We added a paper flower, a paper water container, a nutrition container and a light into the real plant to made it more realistic and made it like the virtual plant.

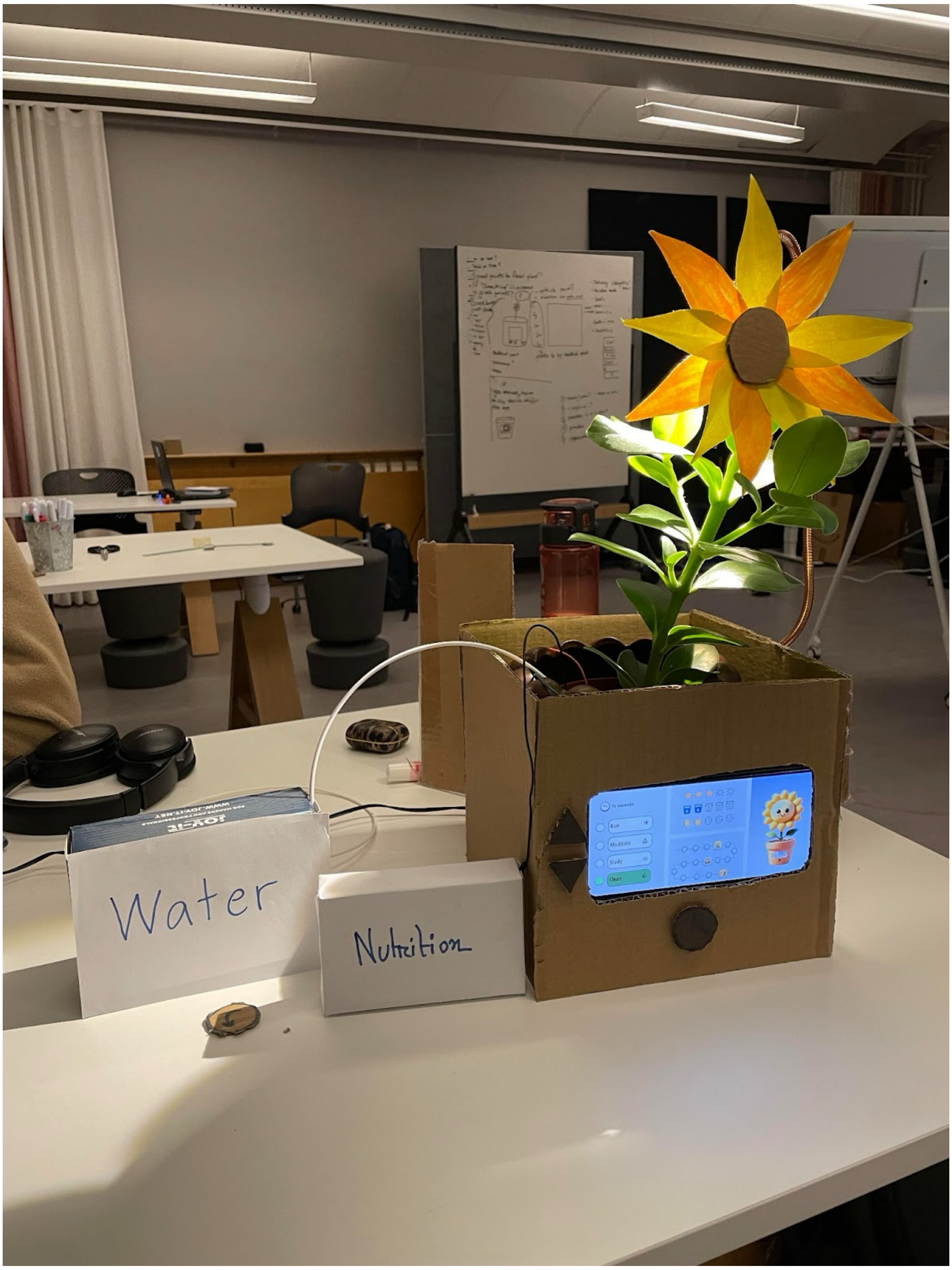

Figure 13: Final prototype of the blossom buddy(high-fidelity)

3.Evaluation of the Final Version of the Blossom-Buddy Prototype:

3.1 User Testing Sessions:

As we used Woz prototyping technique, one of our group members were controlling the light and the screen. Firstly, we gave a scenario for an individual two days and a sad scenario, we analyzed the user reaction and feedback according to those two days. We did not give any hints or overview of the interactive system. We let them use the system and tell them to navigate it slowly and see over the changes.

Figure 14:User test

3.2 Scenario 1(First day):

After finishing a productive day, the user arrived home. He had done all the tasks for example: he ran, meditated, studied, and cleaned up your desk at work and his goal had been achieved for that day. Now he had to report the blossom buddy. When he reported his first task,got a cookie/sun/or water)

3.3 Scenario 2(Second day):

The user just woke up and decided to run. After finishing running the user must report Blossom Buddy. (Here on the second day, there was a special reward to be unlocked and we wanted that reaction from the user.)

3.4 Sad Scenario:

The user just came home and had a very hectic day. Could not do any task he was supposed to do. When he noticed this virtual plant, he saw that the plant was sad because it was running out of its resources. To feed the plant he had to do quick tasks like
do one of these tasks:

- “study” and read this paragraph.
- or clean the desk.
- or meditate for one minute and then report it to Blossom-Buddy.

We have done this user test with several users. At the same time, the team conducted a semi-interview with clear feedback from the user test.

3.5 User Feedback:

The feedback and reactions from the user test were pretty mixed. Almost every user preferred to have only a touch screen instead of physical buttons because they were having a hard time understanding the button selection process. They had a clear understanding of the sun, water, and nutrition bar. They liked the motivational reward system. Two of them loved the hat instead of the background reward. They have suggested some reactions to the flower like being too sad or a little sad. One of our users recommended a toy-like flower instead of the real plant to get an immediate reaction as getting the reaction from the real plant takes a long time. Another User wanted to see his daily progress and recommended we make the progress motivational.
reward line like a calendar. Everyone agreed about the display placement. They thought the placement of the screen was okay. Also, they did not want to band to see or use the screen and wanted to use the tilted screen.

4. Future development:
As this one is the first iteration, we have we are going to use the user's feedback in our future iteration. In the future, we are going to build an app for our display where users can show their daily, weekly, monthly, and yearly improvements. The app would have such a feature that will help the user control the virtual plant display remotely. It will have a vacation mood therefore the user can skip it during his vacation and the plant will automatically take care of itself.

5. Reflection:
The concept was pretty unique to me. I liked the combination of nature and technology. The blossom buddy was very successful in creating emotional engagement with the user. Users took the combination of nature and technology quite positively. The task-based reward system was successfully able to motivate the user to do their task as well as they could easily understand the connection between real plants and virtual plants. I asked to do something differently I would add an undo button as it seemed very important from the beginning. Furthermore, after user testing some simple and quick changes could be made. Additionally, I learned many things such as time management, and how to make different types of prototypes and I had a chance to apply the theories and methodology in a real-life project that I had learned from the course and iterative process of development. However, everything was useful in this system, I barely could find any less useless thing in this interactive system.

6.The point of view of constructive design research:
Constructive design research is a way of research in the design field that does not maintain any traditional ways of designing but rather actively creates new ways of design through the design process (Koskinen et al., 2011). Blossom buddy goes well with this research process. As constructive design research is solution, exploration, and explanation-oriented design research it fits with the blossom buddy concept. Combining nature and technology is not a traditional approach to designing the research process. The Value Proposition plays a special role between constructive design research and the final product. I have created a value proposition canvas to the final version of our 'Blossom Buddy' to encapsulate the unique value the product delivers.

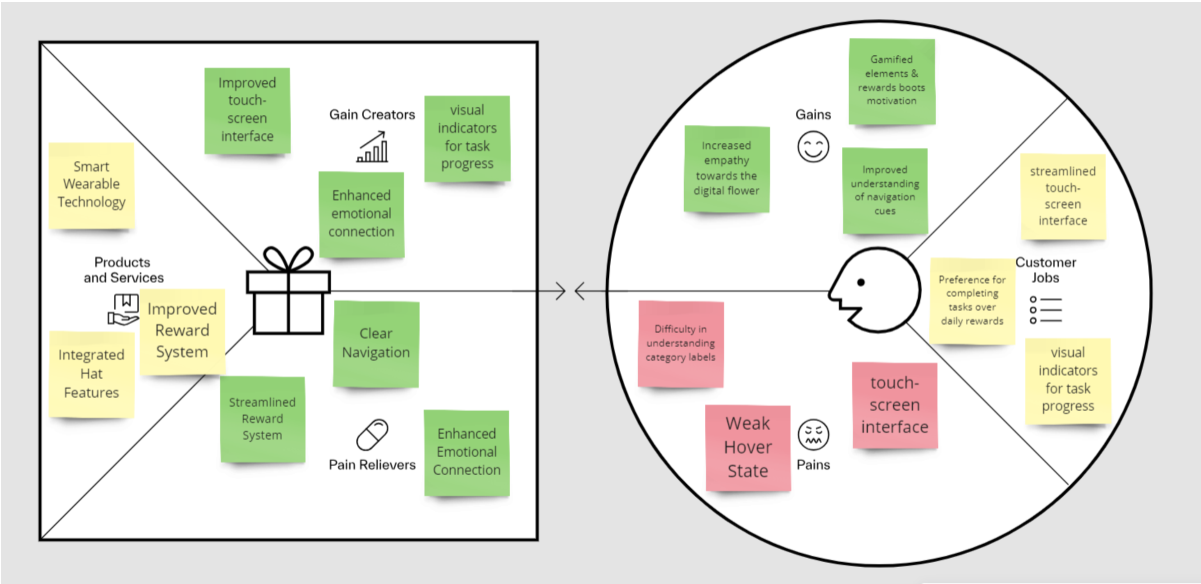

Figure 15: Value Proposition Canvas for Final Version of 'Blossom Buddy'

Conclusion: Blossom Buddy is an innovative motivational interactive system combining nature and technology. With real-plant interaction, a motivational reward approach, and user-centered design, it stands out from other interactive self-development systems.

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